

RIYA KANANI

Ellicott City, MD 21043

443-996-5839 ♦ riya.kananiwork@gmail.com ♦ [LinkedIn](#) ♦ [Website](#) ♦ [Github](#)

EDUCATION

Carnegie Mellon, Master of Entertainment Technology

Exp. May 2027

University of Maryland, Bachelor of Science

May 2025

Computer Science and Immersive Media Design

GPA: 3.85

SKILLS

Software: Unity Engine, Adobe Suite, Microsoft Suite, Cinema4D, Autodesk Fusion, Figma, Mapbox, Forgejo, Bitbucket, OpenCV

Languages: Java, Python, Linux, C, MIPS Assembly, Ruby, Rust, Latex, C#, C++, JS, HTML/CSS, Svelte, HLSL, SQL

WORK EXPERIENCE

Cortina Productions

McLean, VA

Software Engineer Intern

Jun 2025 – Aug 2026

- Built real-time interaction systems in Unity 6 (C#, URP) for four concurrent public installations, holding a 60 FPS budget on always-on hardware; profiled and reduced CPU load roughly 60% → 30% by eliminating per-frame allocations and tightening device polling loops.
- Designed and shipped a novel physical input system — dual UHF RFID readers plus depth-camera hand tracking (Orbbec/Nuitrack, MediaPipe) — translating page turns and mid-air hover into game state, including transition handling for the ambiguous "mid-flip" case so animation never desynced from physical input.
- Made the input layer plug-and-play for non-engineers: auto-discovered hardware by device PID and cached the mapping to config, cutting cold-start device connection from ~25s to under 3s and removing manual COM-port setup entirely.
- Replaced a ~20-minute manual calibration step with a runtime computer-vision routine that edge-detects projected interaction targets on launch — turning a per-install engineering task into zero-touch setup.

University of Maryland

College Park, MD

Teaching Assistant: Introduction to Immersive Media

Aug 2023 – May 2024

- Provided debugging assistance for **Unity Engine** projects and **C#** scripts twice a week in class and office hours.
- Guided students in transforming ideas into successful projects and provided constructive feedback on weekly assignments to support academic and professional growth.

ACTIVITIES

[Girls Who Code](#)

College Park, MD

Web Developer and Teaching Assistant

Feb 2023 – May 2025

- Contributed to the website development and maintenance in **weekly meetings**, utilizing tools including Figma, HTML, and CSS.
- Provided instructions in python fundamentals to middle and high school aged girls twice a semester.

RESEARCH

University of Maryland

College Park, MD

Undergraduate Researcher:

Jun 2024 – May 2025

[SuperFoldAE](#) & [ConSOLAE](#)

- Designed and implemented a supervised autoencoder model with auxiliary reconstruction tasks to enhance model generalization and stability for protein fold prediction, achieving an **88.73% accuracy**.
- Developed a contractive supervised autoencoder architecture that learns smooth, noise-invariant latent spaces for fold and family classification.
- Improved out-of-distribution performance on SCOP 2.06 from **22.63%** → **54.22%** Top-1 accuracy, demonstrating significant advances in generalization
- Leveraged a high-performance computing cluster for efficient model training and data processing.
- Introduced contractive **regularization**, **Smooth L1 reconstruction**, and a **deeper encoder-decoder structure** to stabilize training and capture fine-grained structural features.

NASA Ocean Project

College Park, MD

Researcher

Aug 2023 – May 2024

- Authored C# scripts to simulate phytoplankton responding to camera input data in Unity Engine.
- Played a key role in weekly brainstorming sessions to create an effective method to represent phytoplankton across various zoom scales, ensuring visual cohesion and engaging the user.